MSG No. 187.212 $P\bar{6}'m2'$ [Type III, hexagonal]

Table 1: Wyckoff site: `1a`, site symmetry: `-6'm2'`

No.	position	mapping
1	[0, 0, 0]	`[1,2,3,4,5,6,7,8,9,10,11,12]`

Table 2: Wyckoff site: `1b`, site symmetry: `-6'm2'`

No.	position	mapping
1	$\left[0,\ 0,\ \frac{1}{2}\right]$	`[1,2,3,4,5,6,7,8,9,10,11,12]`

Table 3: Wyckoff site: `1c`, site symmetry: `-6'm2'`

No.	position	mapping
1	$\left[\frac{1}{3},\ \frac{2}{3},\ 0\right]$	`[1,2,3,4,5,6,7,8,9,10,11,12]`

Table 4: Wyckoff site: `1d`, site symmetry: `-6'm2'`

No.	position	mapping
1	$\left[\frac{1}{3},\ \frac{2}{3},\ \frac{1}{2}\right]$	`[1,2,3,4,5,6,7,8,9,10,11,12]`

Table 5: Wyckoff site: `1e`, site symmetry: `-6'm2'`

No.	position	mapping
1	$\left[\frac{2}{3},\ \frac{1}{3},\ 0\right]$	`[1,2,3,4,5,6,7,8,9,10,11,12]`

Table 6: Wyckoff site: `1f`, site symmetry: `-6'm2'`

No.	position	mapping
1	$\left[\frac{2}{3},\ \frac{1}{3},\ \frac{1}{2}\right]$	`[1,2,3,4,5,6,7,8,9,10,11,12]`

Table 7: Wyckoff site: `2g`, site symmetry: `3m.`

No.	position	mapping
1	$[0,\ 0,\ z]$	`[1,2,3,4,5,6]`
2	$[0,\ 0,\ -z]$	`[7,8,9,10,11,12]`

Table 8: Wyckoff site: `2h`, site symmetry: `3m.`

No.	position	mapping
1	$\left[\frac{1}{3},\ \frac{2}{3},\ z\right]$	`[1,2,3,4,5,6]`
2	$\left[\frac{1}{3},\ \frac{2}{3},\ -z\right]$	`[7,8,9,10,11,12]`

Table 9: Wyckoff site: `2i`, site symmetry: `3m.`

No.	position	mapping
1	$\left[\frac{2}{3},\ \frac{1}{3},\ z\right]$	`[1,2,3,4,5,6]`
2	$\left[\frac{2}{3},\ \frac{1}{3},\ -z\right]$	`[7,8,9,10,11,12]`

Table 10: Wyckoff site: `3j`, site symmetry: `m'm2'`

No.	position	mapping
1	$[x,\ -x,\ 0]$	`[1,5,9,11]`
2	$[x,\ 2x,\ 0]$	`[2,6,7,12]`
3	$[-2x,\ -x,\ 0]$	`[3,4,8,10]`

Table 11: Wyckoff site: `3k`, site symmetry: `m'm2'`

No.	position	mapping
1	$\left[x,\ -x,\ \frac{1}{2}\right]$	`[1,5,9,11]`
2	$\left[x,\ 2x,\ \frac{1}{2}\right]$	`[2,6,7,12]`
3	$\left[-2x,\ -x,\ \frac{1}{2}\right]$	`[3,4,8,10]`

Table 12: Wyckoff site: `6l`, site symmetry: `m'..`

No.	position	mapping
1	$[x,\ y,\ 0]$	`[1,11]`
2	$[-y,\ x-y,\ 0]$	`[2,12]`
3	$[-x+y,\ -x,\ 0]$	`[3,10]`
4	$[-x+y,\ y,\ 0]$	`[4,8]`
5	$[-y,\ -x,\ 0]$	`[5,9]`
6	$[x,\ x-y,\ 0]$	`[6,7]`

Table 13: Wyckoff site: `6m`, site symmetry: `m'..`

No.	position	mapping
1	$\left[x,\ y,\ \frac{1}{2}\right]$	`[1,11]`
2	$\left[-y,\ x-y,\ \frac{1}{2}\right]$	`[2,12]`
3	$\left[-x+y,\ -x,\ \frac{1}{2}\right]$	`[3,10]`
4	$\left[-x+y,\ y,\ \frac{1}{2}\right]$	`[4,8]`
5	$\left[-y,\ -x,\ \frac{1}{2}\right]$	`[5,9]`

continued ...

Table 13

No.	position	mapping
6	$\left[x,\, x-y,\, \frac{1}{2}\right]$	`[6,7]`

Table 14: Wyckoff site: `6n`, site symmetry: `.m.`

No.	position	mapping
1	$[x,\, -x,\, z]$	`[1,5]`
2	$[x,\, 2x,\, z]$	`[2,6]`
3	$[-2x,\, -x,\, z]$	`[3,4]`
4	$[x,\, 2x,\, -z]$	`[7,12]`
5	$[-2x,\, -x,\, -z]$	`[8,10]`
6	$[x,\, -x,\, -z]$	`[9,11]`

Table 15: Wyckoff site: `12o`, site symmetry: `1`

No.	position	mapping
1	$[x,\, y,\, z]$	`[1]`
2	$[-y,\, x-y,\, z]$	`[2]`
3	$[-x+y,\, -x,\, z]$	`[3]`
4	$[-x+y,\, y,\, z]$	`[4]`
5	$[-y,\, -x,\, z]$	`[5]`
6	$[x,\, x-y,\, z]$	`[6]`
7	$[x,\, x-y,\, -z]$	`[7]`
8	$[-x+y,\, y,\, -z]$	`[8]`
9	$[-y,\, -x,\, -z]$	`[9]`
10	$[-x+y,\, -x,\, -z]$	`[10]`
11	$[x,\, y,\, -z]$	`[11]`
12	$[-y,\, x-y,\, -z]$	`[12]`